

GoHa Automations

Making the system reach you, without putting any of it inside the app.

Phase-by-phase build guide · Make.com, n8n, and native iPhone

GoHa deliberately contains *no* notification infrastructure. That is written into the project constitution: push, email, Telegram and third-party automation platforms are out of scope, and external automations are to be **layered on top, never baked in**. Everything in this guide sits outside the app and reads the same database the app reads.

[Neon Postgres](#) → [n8n](#) / [Make](#) → [push service](#) → [iPhone](#)

Answering your question first: iPhone, not Telegram

Yes. You do not need Telegram, and on iOS you have better options. There are two directions and they need different tools.

Direction	Best tool	Why
Push server → your phone	Pushcut (recommended) or Pushover	Gives you a webhook URL. Post JSON to it and a real iOS notification appears. Pushcut's notifications can carry action buttons that run Apple Shortcuts , which is what makes two-way automation possible later. Pushover is simpler and cheaper if you only ever want to be told things.
Pull you → the system	Apple Shortcuts + Scriptable	Native, free, already on your phone. Handles Siri ("what should I focus on?"), Home Screen and Lock Screen widgets, Focus-mode triggers, and Back Tap.

Apple Shortcuts cannot receive a push. A Personal Automation can run *on a schedule* and fetch a URL, but nothing external can wake it. That is the one hard limit, and it is why a push service (Pushcut/Pushover) is needed for anything server-initiated. Shortcuts alone can only poll on a timer, which drains battery and misses anything time-sensitive.

What each tool is actually for

Tool	Cost	Use it for
Make.com	Free tier	Simple scheduled notifications. Visual, quick to build, nothing to host.
n8n	Free self-host or cloud	Anything with real logic: multi-step conditions, loops, JavaScript, deduplication, state. This is where the "smart" automations live.
Pushcut	Free tier	The bridge that turns a webhook into an iPhone notification with buttons.
Shortcuts	Free	Widgets, Siri, Focus triggers, and the actions your notification buttons run.
Scriptable	Free	A Home Screen widget that renders today's brief.

An honest note on Make vs n8n. n8n can do everything Make can. Running both means two bills, two places to debug, and two copies of your credentials. Use Make for Phases 1–2 if the visual builder helps you get started, but once n8n is up in Phase 3, consider moving the simple flows across and retiring Make. One system you know well beats two you half-know.

Phase 0 — Foundations

Do this before any automation. It is thirty minutes that prevents every later security problem.

0.1 A read-only database role

DATABASE

Never give an automation your `neondb_owner` string. Create a role that can read and nothing else, so a leaked credential cannot delete a year of history.

```
-- Run once in the Neon SQL editor, as the owner
CREATE ROLE goha_automation WITH LOGIN PASSWORD '<a-long-random-password>';
GRANT CONNECT ON DATABASE neondb TO goha_automation;
GRANT USAGE ON SCHEMA public TO goha_automation;
GRANT SELECT ON ALL TABLES IN SCHEMA public TO goha_automation;
-- Cover tables you add later
ALTER DEFAULT PRIVILEGES IN SCHEMA public
  GRANT SELECT ON TABLES TO goha_automation;
```

Build a connection string for this role from your Neon dashboard and store it as a credential inside n8n/Make. Never paste it into a node body.

0.2 Find your user id

DATABASE

Every GoHa table is user-scoped. Every query you write needs your id.

```
SELECT id, email FROM "user" WHERE email = '<your-email>';
```

Store the UUID as an environment variable (`GOHA_USER_ID`), not inline in each query.

0.3 Set up the push channel and test it

IPHONE

1. Install **Pushcut** from the App Store and allow notifications.
2. In Pushcut, create a notification named `goha`. Copy its webhook URL.
3. Test it before building anything, from any terminal:

```
curl -X POST "https://api.pushcut.io/<secret>/notifications/goha" \
  -H "Content-Type: application/json" \
  -d '{"title":"GoHa","text":"Wiring works."}'
```

If that buzzes your phone, the hardest part is already done.

0.4 Decide your quiet rules now

POLICY

Write these down before you build, because every automation must obey them: **never notify when there is nothing to act on**, **never send the same thing twice in a day**, and **nothing between 22:00 and 06:00**. A daily "you have 4 tasks" becomes wallpaper within a week, and a muted automation is worse than none.

Phase 1 — The morning brief

One notification a day that tells you what to start with. This is the whole payoff; everything after it is refinement.

1 Scheduled brief

MAKE.COM

Schedule 07:30 → Postgres query → Router (empty?) → Pushcut

Modules

1. **Schedule** — daily at your planning time. Set the scenario timezone to `Asia/Manila` or every date comparison will be a day out.
2. **PostgreSQL** → **Execute a query** using the read-only role.
3. **Router / Filter** — stop if the query returned no rows.
4. **HTTP** → **Make a request** — POST to the Pushcut webhook.

The query

This mirrors how Today ranks work: lateness first, then priority.

```
SELECT t.title, t.priority,
       (CURRENT_DATE AT TIME ZONE 'Asia/Manila')::date
       - COALESCE(t.scheduled_for, (t.due_at AT TIME ZONE 'Asia/Manila')::date)
       AS days_late
FROM tasks t
WHERE t.user_id = '<GOHA_USER_ID>'
      AND t.parent_task_id IS NULL
      AND t.status IN ('todo', 'in_progress')
      AND COALESCE(t.scheduled_for, (t.due_at AT TIME ZONE 'Asia/Manila')::date)
      <= (now() AT TIME ZONE 'Asia/Manila')::date
ORDER BY days_late DESC,
       CASE t.priority WHEN 'urgent' THEN 3 WHEN 'high' THEN 2
                       WHEN 'medium' THEN 1 ELSE 0 END DESC
LIMIT 3;
```

Notification body: "Start with: `{{title}}`", and when `days_late > 0`, append " — `{{days_late}}` days late".

2 Evening check, but only if it matters

MAKE.COM

Runs at your reflection time. Sends **nothing** unless a habit is unchecked or the day closed with zero completions. Silence is the feature.

```
-- habits due today with no entry yet
SELECT h.name
FROM habits h
JOIN habit_schedules s ON s.habit_id = h.id AND s.is_active
WHERE h.user_id = '<GOHA_USER_ID>'
      AND NOT h.is_archived
      AND (s.start_date IS NULL OR s.start_date <= CURRENT_DATE)
      AND NOT EXISTS (
        SELECT 1 FROM habit_entries e
        WHERE e.habit_id = h.id
              AND e.entry_date = (now() AT TIME ZONE 'Asia/Manila')::date
      );
```

Phase 2 — Streak rescue

The single highest-value automation you will build. It saves something real.

3 Protect a streak before it breaks

[MAKE.COM](#)

At 20:00, find habits with a streak of 3 or more days that have **no entry today**. Those are the only ones worth interrupting your evening for.

```
-- consecutive days completed up to yesterday, per habit
WITH done AS (
  SELECT habit_id, entry_date,
         entry_date - (ROW_NUMBER() OVER (
           PARTITION BY habit_id ORDER BY entry_date))::int AS grp
  FROM habit_entries
  WHERE user_id = '<GOHA_USER_ID>' AND status = 'done'
),
runs AS (
  SELECT habit_id, COUNT(*) AS len, MAX(entry_date) AS last_day
  FROM done GROUP BY habit_id, grp
)
SELECT h.name, r.len AS streak
FROM runs r
JOIN habits h ON h.id = r.habit_id
WHERE r.last_day = (now() AT TIME ZONE 'Asia/Manila')::date - 1
  AND r.len >= 3
  AND NOT h.is_archived;
```

Message: "{{streak}}-day streak on {{name}}. Still time." Nothing else. No list, no summary.

Phase 3 — n8n and a proper API

Where the smart automations begin. This phase builds no notifications; it builds the foundation the next ones stand on.

4 Stand up n8n

[N8N](#)

Self-host with Docker, or use n8n Cloud if you would rather not run anything.

```
docker run -d --name n8n --restart unless-stopped \
  -p 5678:5678 \
  -e GENERIC_TIMEZONE="Asia/Manila" \
  -e TZ="Asia/Manila" \
  -e N8N_ENCRYPTION_KEY="<a-long-random-key>" \
  -v n8n_data:/home/node/.n8n \
  docker.n8n.io/n8nio/n8n
```

Add two credentials immediately: the read-only Postgres role, and the Pushcut URL as a header-auth credential. Set the timezone at the container level so every Cron node agrees with your data.

5 A read API on GoHa itself

GOHA

SQL can find late tasks, but it cannot reproduce Today's judgement. That logic lives in `lib/today-brain.ts`, and it is **pure and importable** — reimplementing it in SQL would guarantee the two drift apart.

Add one route handler that runs the real engine and returns its output:

```
// app/api/automation/brief/route.ts (sketch)
// Auth: a bearer token from env, compared in constant time.
// Then: load the same records Today loads, call deriveDaySignal(),
// and return { headline, detail, task, lateCount, suggestions }.
```

Rules this endpoint must keep. Bearer token in an `Authorization` header, never a query string (query strings end up in logs). Compare it with a timing-safe equality check. Return only your own data, derived from the token, never from a caller-supplied `user_id`. Read-only. Rate limit it. This is the one place the app grows a public surface, so it is worth being strict.

6 Rebuild the brief on n8n

N8N

Cron → HTTP (GoHa API) → IF → Function (format) → HTTP (Pushcut)

Now the notification says exactly what the app says, because it is the same code. When you improve the ranking in GoHa, the notification improves with it and you change nothing here.

Phase 4 — The smart automations

Things the app cannot do for itself because they need to run when you are not looking at it.

7 Goal at risk

N8N

Weekly. Find active goals whose target date is close but whose progress is not, and say so while there is still time to act.

```
SELECT g.title, g.target_date,
       COUNT(t.*) FILTER (WHERE t.status = 'completed') AS done,
       COUNT(t.*) FILTER (WHERE t.status <> 'cancelled') AS total
FROM goals g
LEFT JOIN tasks t ON t.goal_id = g.id
WHERE g.user_id = '<GOHA_USER_ID>'
      AND g.status = 'active' AND NOT g.is_archived
      AND g.target_date IS NOT NULL
      AND g.target_date < CURRENT_DATE + 21
GROUP BY g.id;
```

In a Function node, flag a goal when `done/total` is below the fraction of its timeline already elapsed. That is a genuine early warning, not a countdown.

8 Neglected life area

N8N

Monthly. For each life area, count tasks completed in the last 30 days. If an area you rated important (`life_areas.weight >= 4`) has none, that is worth knowing: it is exactly the drift a life-balance system exists to catch, and nothing in the UI surfaces it today.

9 Overdue escalation, with backoff

N8N

Anything **3+ days late**, notified **once**, then not again for a week. Store what you have already sent (n8n static data, or a small table in your own database) and check it before sending. Without backoff this becomes the automation you mute, and then you mute all of them.

10 Sunday review nudge

N8N

Sunday evening, only if this week has no completed review:

```
SELECT 1 FROM weekly_reviews
WHERE user_id = '<GOHA_USER_ID>'
      AND week_start = date_trunc('week', (now() AT TIME ZONE 'Asia/Manila'))::date
      AND completed_at IS NOT NULL;
```

If no row, send a brief with the week's numbers and a deep link. Note this assumes a Monday week start; match it to your Settings preference.

11 Nightly backup

N8N

Not a notification, and the most important one here. Nightly, call GoHa's export endpoint (or `pg_dump` via the read-only role), write the file to cloud storage, and notify **only on failure**. A single hosted database with no backup is the real risk in this whole system.

Phase 5 — Two-way: acting from the notification

Where it stops being a reminder service and starts being a control surface.

12 Action buttons

IPHONE

A Pushcut notification can carry buttons. Each button calls a URL, which hits an n8n webhook, which calls a **write** endpoint on GoHa.

Notification → button → n8n webhook → GoHa API → database

Buttons worth having: **Done**, **Snooze to tomorrow**, **Pin to Top 3**, **Start focus**.

Writes must go through GoHa, never straight into Postgres. The app enforces Zod validation, ownership checks and revalidation on every mutation. An automation writing directly to the database bypasses all of it and can produce rows the app considers impossible. Keep the automation role `SELECT` -only, so this stays true by construction rather than by discipline.

Phase 6 — Native iPhone surfaces

No server involved. Pure Apple, and the part you will use most often.

13 Home Screen widget

SCRIPTABLE

A Scriptable widget that calls your brief endpoint and renders the headline and top task. Refreshes on its own. This ends up replacing most notifications, because you see it every time you unlock your phone and it never interrupts you.

14 Siri and Back Tap

SHORTCUTS

A Shortcut named **"What's my focus?"** that fetches the brief and speaks it. Then say "Hey Siri, what's my focus?" — hands-free, in the car, before you have opened anything. Assign the same Shortcut to Back Tap (Settings → Accessibility → Touch → Back Tap) for a double-tap on the back of the phone.

15 Focus mode triggers

SHORTCUTS

A Personal Automation on *"When Work Focus turns on"* that starts a GoHa focus session on your top task, and on *"turns off"* that ends it. Your phone's Focus mode and the app's Focus Mode become the same act.

16 Calendar subscription

GOHA

If you want GoHa's scheduled work inside the native Calendar app, the clean way is an `.ics` feed endpoint on a secret URL, subscribed to in iOS Settings → Calendar → Accounts → Add Subscribed Calendar. One caveat worth stating plainly: this adds a **calendar surface to the app**, and calendar sync is listed as out of scope in your constitution. A read-only feed is a defensible reading of that rule, but it is your call, not something to add by accident.

Build order

Order	Build	Where	Why here
1	Read-only role, user id, Pushcut test	Neon + iPhone	Nothing works safely without it
2	Morning brief	Make.com	The whole payoff, in one flow
3	Streak rescue	Make.com	Saves something real, immediately
4	Nightly backup	n8n	Removes your biggest actual risk
5	GoHa brief API + n8n	GoHa + n8n	Notifications start matching the app exactly
6	Scriptable widget	iPhone	Replaces most notifications with a glance
7	Goal at risk, neglected area, review nudge	n8n	The genuinely smart layer
8	Action buttons and writes	All	Only once the read side is trustworthy

Do Phase 0 and one automation this week. Build the morning brief, live with it for seven days, and see whether you actually read it. That answer should decide what you build next far more than this list does.

Operating rules

- **Timezone everywhere.** Set `Asia/Manila` on the n8n container, the Make scenario, and inside every query. A UTC default silently shifts your "today" by eight hours.
- **Idempotency.** Assume every flow can run twice. Key each notification on `(kind, date)` and skip if already sent.
- **Fail loudly, once.** An error workflow that notifies you when an automation breaks, with backoff. Silent failure is worse than no automation, because you will trust it anyway.
- **Secrets stay in the credential store.** Never in a node body, a query string, or a screenshot.
- **Log what you send.** A small table of sent notifications makes deduplication trivial and gives you something to review when you start ignoring them.
- **Review the automations monthly.** Delete any you have started swiping away. That is data about what you actually want, and it is worth more than the automation was.

What not to automate

- **Creating tasks for you.** A system that fills its own inbox stops being trusted.
- **Auto-rescheduling overdue work.** Moving a date is a decision. Automating it removes the friction that makes you notice you overcommitted.
- **Auto-completing habits.** The value of a streak is that it is honest.
- **Anything with an AI summary of your day.** Your constitution rules it out, and Today's brain already gives you a stated, inspectable reason instead of a generated one.

GoHa automations sit entirely outside the app. GoHa itself keeps no notification infrastructure, no scheduler and no third-party integrations, per CLAUDE.md section 2 — the only additions on the app side are optional, token-authenticated, read-mostly endpoints.